

LESSON 8.4

SOLUTIONS TO LINEAR EQUATIONS



LESSON INTRODUCTION

MA.8.AR.2.1 Solve multi-step linear equations in one variable with rational number coefficients. Include equations with variables on both sides.

LEARNING TARGETS

- I can solve equations with variables on both sides.
- I can identify when an equation has infinitely many solutions or no solution.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

WORKING TOWARD

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

MA.912.AR.2.1 Given a real-world context, write and solve one-variable, multi-step linear equations.

ESSENTIAL QUESTIONS

- How do you solve an equation with variables on both sides?
- How does the structure of an equation relate to the number of solutions it has?

LESSON NARRATIVE

This lesson builds on students' previous experiences solving equations with variables on both sides of the equals sign. Students will continue solving equations in one variable but will also extend their knowledge to consider the solutions an equation has based on its structure. Students will explore equations that have an infinite number of solutions and no solution, in addition to working with equations with one solution. Students will learn to use the structure of an equation to determine the number of solutions it has.

COMPONENT SUMMARY

8.4.1 WARM-UP (~5 MIN.)

Use an equation in perimeter context

8.4.3 EXPLORATION (10-15 MIN.)

Explore the number of solutions to equations

8.4.5 WRAP-UP (~5 MIN.)

Reflect on learning throughout lesson

8.4.2 COLLABORATION (10-15 MIN.)

Practice solving multi-step equations

8.4.4 GUIDED PRACTICE (10 MIN.)

Use the structure of an equation to determine the number of solutions

RESOURCES

GLOSSARY

Key Terms:

- Infinitely many solutions
- No solution

Additional Terms: N/A

MATERIALS

- (Optional) Scratch paper or chart paper
- (Optional) Scientific calculator

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to understand the concept of infinitely many solutions.
- Students may struggle to understand that an equation may have more than one solution or no solution and try to assign a solution to the problem.
- Students may incorrectly think that an equation with a solution of 0 has no solution.

THINGS TO CONSIDER

- As students encounter equations with one solution, no solution, and infinitely many solutions, consider constructing an anchor chart with three columns of examples so they can begin to better observe the structure of the equations as they move through the lesson.
- Allowing students to use calculators during the 8.4.3 Exploration and 8.4.4 Guided Practice will help students focus on discovery and patterns in equations rather than performing calculations. Consider also allowing students to use their calculator to verify their solutions in the 8.4.2 Collaboration.

LITERACY AND LANGUAGE ROUTINES

Collect and Display

Listen for, collect, and display vocabulary that students use to discuss the equations and types of solutions. Throughout the lesson, continue to update collected student language and remind students to borrow language from the display as needed. This will help students use mathematical language during their paired and whole-group discussions.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

Students will be working to describe how the structure of an equation contributes to the number of solutions it has. Clarifications within this standard emphasize recognizing patterns and applying these patterns to mathematical concepts. Utilize the progression of the lesson to highlight the process by which students use features of equations to form generalizations about the number of solution(s).

DIFFERENTIATE INSTRUCTION

Provide numerous ways for students to provide feedback, including written and oral methods. If students feel overwhelmed by the number of problems, consider allowing students to solve fewer of them. Consider allowing students to work problems on scratch paper or chart paper, if they struggle to write the problems in smaller spaces.

8.4.1 WARM-UP: BELL WORK

ENRICHMENT

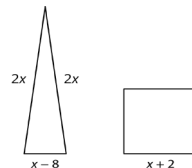
If time allows, consider allowing students to discuss their answers to these questions with a partner after completing 8.4.1 Warm-Up:

- Which different methods did you use to solve for x ?
- How did you use the solution to find the perimeter of the triangle and the square?



8.4.1 Warm-Up: Bell Work

A triangle and a square are shown with their dimensions given in terms of x units. The triangle and square have the same perimeter.



- The equation $4(x + 2) = 2x + 2x + x - 8$ can be used to find the value of x based on the information given.

- Complete the table to indicate which figure's perimeter each expression represents.

Expression	Perimeter of the ...
$2x + 2x + x - 8$	<i>Triangle</i>
$4(x + 2)$	<i>Square</i>

- Find the value of x .

$$4(x + 2) = 2x + 2x + x - 8$$

$$4x + 8 = 5x - 8$$

$$x = 16$$

- What is the perimeter of each figure?

$$\text{Triangle: } 2(16) + 2(16) + (16) - 8 = 72 \text{ units}$$

$$\text{Square: } 4(16 + 2) = 72 \text{ units}$$



8.4.2 Collaboration: Which Would You Prefer?

Nine equations are shown.

$-\frac{1}{2}(t + 3) - 10 = -6.5$	$\frac{10-v}{4} = 2(v + 17)$
$\frac{n}{7} - 12 = 5n + 5$	$\frac{4m-3}{4} = -\frac{9+4m}{8}$
$2r + 49 = -8(-r - 5)$	$p - 5(p + 4) = p - (8 - p)$
$2(2q + 1.5) = 18 - q$	$2(4k + 3) - 13 = 2(18 - k) - 13$
$3(c - 1) + 2(3c + 1) = -(3c + 1)$	

1. Consider the equations with your partner.
- a. Without solving any of the equations, identify three equations that you and your partner think would be least difficult to solve and write an L in each box next to these equations.
Answers will vary.
 - b. Then, identify three equations you think would be the most difficult to solve and write an M in each box next to these equations.
Answers will vary.

2. Choose three equations to solve. One equation must be from your list of least difficult equations, one from your list of most difficult, and one that was not on either list.
- a. Circle the equations you will be solving from the original table.
Answers will vary.
 - b. Work with your partner to solve each equation. Write the equation being solved in the table, and show your work to find each solution in the space provided.
Answers will vary. See solutions below.

List	Equation	Work and Solution
Least Difficult	<i>Answers will vary.</i>	<i>Answers will vary.</i>
Most Difficult	<i>Answers will vary.</i>	<i>Answers will vary.</i>
Neither	<i>Answers will vary.</i>	<i>Answers will vary.</i>

$-\frac{1}{2}(t + 3) - 10 = -6.5$ $t = -10$	$\frac{10-v}{4} = 2(v + 17)$ $v = -14$
$\frac{n}{7} - 12 = 5n + 5$ $n = -3.5$	$\frac{4m-3}{4} = -\frac{9+4m}{8}$ $m = -\frac{1}{4}$
$2r + 49 = -8(-r - 5)$ $r = \frac{3}{2}$	$p - 5(p + 4) = p - (8 - p)$ $p = -2$
$2(2q + 1.5) = 18 - q$ $q = 3$	$2(4k + 3) - 13 = 2(18 - k) - 13$ $k = 3$
$3(c - 1) + 2(3c + 1) = -(3c + 1)$ $c = 0$	

8.4.2 COLLABORATION: WHICH WOULD YOU PREFER?

TEACHER MOVES

Observe and listen as students’ make selections in question 1 so they do not change their choices once they read question 2. As students are discussing what they might find challenging, encourage a growth mindset using the power of “yet,” and discuss what strategies they can use to solve the challenging problems.

QUESTIONING

If partners finish early, they can each select another equation to solve independently.

8.4.3 EXPLORATION: HOW MANY SOLUTIONS?

TEACHER MOVES

Allow students to explore these concepts with their partner. There will be an opportunity to solidify understanding in the 8.4.4 Guided Practice.

Observe and listen as students work to determine student thinking to highlight or misconceptions to address once students have completed the exploration with their partners.

TEACHER MOVES

Consider allowing students to use a scientific calculator to verify which values in the table satisfy the equations. The purpose of this task is to consider equations with one or many solutions and make observations about their structure, not necessarily on the computations.

Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator.

TEACHER MOVES

Collect and Display

Circulate throughout the classroom and listen to student pairs discuss the equations and how to create an equation that is true for all values and an equation that is false for all values. Have students display the language and vocabulary that you hear that encourages speaking mathematically, and use the display as a reference during their discussions.

MA.K12.MTR.5.1: Patterns and Structure

Encourage students to use the structure of the equations in the examples from question 1 to determine possible generalizations about what equations that are true for all values and equations that are false for values have in common.



8.4.3 Exploration: How Many Solutions?

Work with your partner to complete the following.

1. The table shows three equations and five possible values of n .
 - a. Evaluate each equation for the values of n and indicate if the value makes the equation true or false.

Equation	$n = -10$	$n = -5$	$n = 0$	$n = 5$	$n = 10$
$n = n$	☐ true ☐ false	☐ true ☐ false	☐ true ☐ false	☐ true ☐ false	☐ true ☐ false
$3n + 2 = 3n + 1$	☐ true ☐ false	☐ true ☐ false	☐ true ☐ false	☐ true ☐ false	☐ true ☐ false
$3(n - 2) = 3n - 6$	☐ true ☐ false	☐ true ☐ false	☐ true ☐ false	☐ true ☐ false	☐ true ☐ false

- b. What do you notice about the structure of the equations that were true for all values of n ?
Answers will vary. The equations have equivalent expressions on both sides of the equation.
2. One of the equations was false for all values of n .
 - a. What do you notice about the structure of the equation that was false for all values of n ?
Answers will vary. The expressions on either side of the equal sign are not equivalent. While both sides of the equation have the same coefficient for n , the constant terms are not the same.

- b. Solve the equation to determine if there is a different value of n that makes the equation true. If there is a solution to the equation, show it. If there is not a solution to the equation, work with your partner to write an explanation for why not.

$$3n + 2 = 3n + 1$$

$$3n + 2 - 3n = 3n + 1 - 3n$$

$$2 \neq 1$$
There is no value for n that could make the statement true because when solving the equation, the result is a false mathematical statement. This equation has no solution.

3. Work with your partner to create an equation that is true for all values.
Answers will vary. Any equation with equivalent expressions on either side will be correct.
4. Work with your partner to create an equation that is false for all values.
Answers will vary. Any equation with the same variable term but different constant terms will be correct.



If an equation is true for all values of the variable, it has **infinitely many solutions**. If there are no values of the variable that make an equation true, it has **no solution**.



8.4.4 Guided Practice: How Many Solutions?

1. For each equation, simplify each side of the equation by either combining like terms or applying the distributive property in the first column. Then, use the rewritten form of the equation to determine if it has one solution, no solution, or infinitely many solutions.

Equation	One Solution	No Solution	Infinitely Many Solutions
$5(b + 3) = -9 + 7b$ $5b + 15 = -9 + 7b$	x		
$2(3c + 3) = 3(2c + 2)$ $6c + 6 = 6c + 6$			x
$8(13 + r) = 48 + 8r$ $104 + 8r = 48 + 8r$		x	
$3a - 5 = 9 + 3a - 14$ $3a - 5 = 3a - 5$			x
$-12s + 14 = -2(6s + 7)$ $-12s + 14 = -12s - 14$		x	

8.4.4 GUIDED PRACTICE: HOW MANY SOLUTIONS?

DIFFERENTIATE INSTRUCTION

Use observational data from 8.4.3 Exploration to determine how much guidance is needed to bring together key ideas in this Guided Practice component. If necessary, consider modeling through a Think-Aloud for the first three equations and then releasing students with their partners for the last two.

TEACHER MOVES

MA.K12.MTR.5.1: Patterns and Structure

Encourage students to make their decision without solving the entire equation. Emphasize and model how to use the structure of an equation to draw generalizations about the type of solution it may have.



8.4.5 Wrap-Up: Reflection

1. Complete the table as you reflect on your work in today's lesson.

Today I deserve 	
for ...	<i>Answers will vary.</i>
Today I 	
by doing ...	<i>Answers will vary.</i>
Today I was 	
by ...	<i>Answers will vary.</i>

8.4.5 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider allowing students to respond using different methods. Some students may prefer reflecting individually, while others may benefit from discussing their learning in a small group.

DIFFERENTIATE INSTRUCTION

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.